

The Riemann Hypothesis

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The Riemann Hypothesis

The Riemann Hypothesis: Formal Statement

The Riemann Hypothesis states:

Every nontrivial zero of the Riemann zeta function

$$\zeta(s)$$

has real part equal to

$$\frac{1}{2}.$$

That is, if

$$\zeta(\rho) = 0 \quad \text{and} \quad 0 < \Re(\rho) < 1,$$

then necessarily

$$\Re(\rho) = \frac{1}{2}.$$

Thus every nontrivial zero lies on the critical line:

$$\Re(s) = \frac{1}{2}.$$

Critical Strip and Critical Line

The nontrivial zeros lie inside the critical strip:

$$0 < \Re(s) < 1.$$

The line

$$\Re(s) = \frac{1}{2}$$

is called the critical line.

Equivalent Statements of RH

The following are equivalent to RH:

1. All nontrivial zeros satisfy

$$\beta = \frac{1}{2}.$$

2. Prime counting error bound:

$$\pi(x) = \text{Li}(x) + O(\sqrt{x} \log x).$$

3. Chebyshev estimate:

$$\psi(x) = x + O(\sqrt{x} \log^2 x).$$

4. Robin's Criterion:

For every integer $n > 5040$,

$$\sigma(n) < e^\gamma n \log \log n.$$

5. Lagarias Criterion:

For every integer $n \geq 1$,

$$\sigma(n) \leq H_n + e^{H_n} \log(H_n),$$

where

$$H_n = \sum_{k=1}^n \frac{1}{k}.$$

6. Mertens function criterion:

If

$$M(x) = \sum_{n \leq x} \mu(n),$$

then RH is equivalent to

$$M(x) = O(x^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0.$$

Why RH Matters

The Riemann Hypothesis gives the sharpest possible estimate for the distribution of prime numbers.

If RH is true, prime numbers are distributed almost as regularly as possible.

It affects:

- Prime number theory
- Cryptography
- Analytic number theory
- Random matrix theory
- Mathematical physics

Riemann Zeta Function

The zeta function is defined by:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \Re(s) > 1.$$

Euler also showed:

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}.$$

This connects zeta zeros directly to prime numbers.